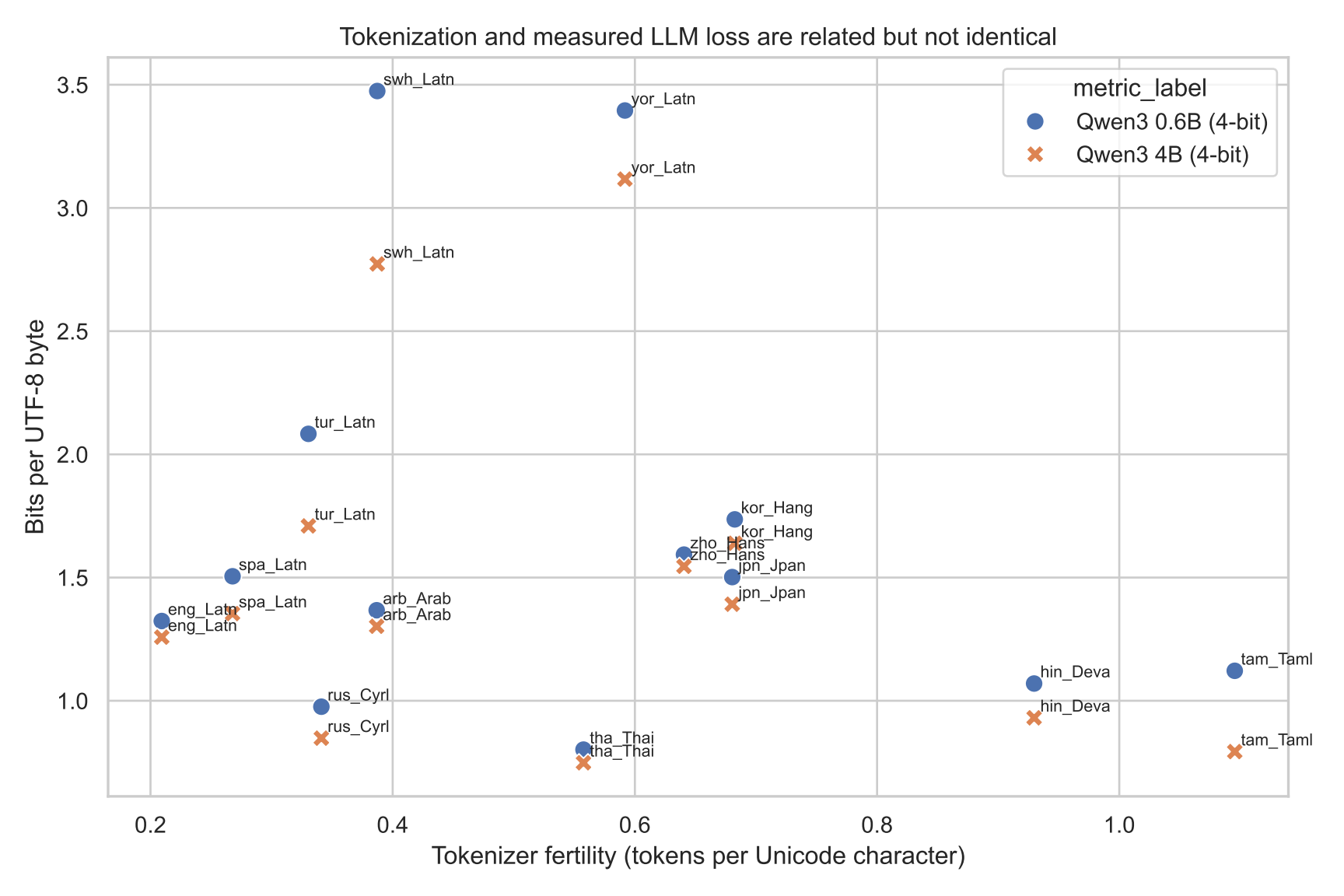
Tokenization and measured LLM loss are related but not identical
metric_label
Qwen3 0.6B (4-bit)
Qwen3 4B (4-bit)
Bits per UTF-8 byte
Tokenizer fertility (tokens per Unicode character)
3.5
3.0
2.5
2.0
1.5
1.0
0.2
0.4
0.6
0.8
1.0
swh_Latn
yor_Latn
yor_Latn
swh_Latn
tur_Latn
tur_Latn
kor_Hang
kor_Hang
zho_Hans
zho_Hans
jpn_Jpan
jpn_Jpan
spa_Latn
spa_Latn
eng_Latn
eng_Latn
arb_Arab
arb_Arab
rus_Cyrl
rus_Cyrl
tha_Thai
tha_Thai
hin_Deva
hin_Deva
tam_Taml
tam_Taml